Lecture 34

Section 6.5

Least-Squares Problems

Let A be a $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Sometimes, there does not exist $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$, i.e. the system is inconsistent. Therefore, we may want to find a vector $\mathbf{x}$ such that $A\mathbf{x}$ is close to $\mathbf{b}$, i.e. a *best approximation*.

Definition: The general least-squares problem is to find an $\mathbf{x}$ that makes $||\mathbf{b} - A\mathbf{x}||$ as small as possible.

Definition: If A is an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$, a least-squares solution of $A\mathbf{x} = \mathbf{b}$ is a $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that

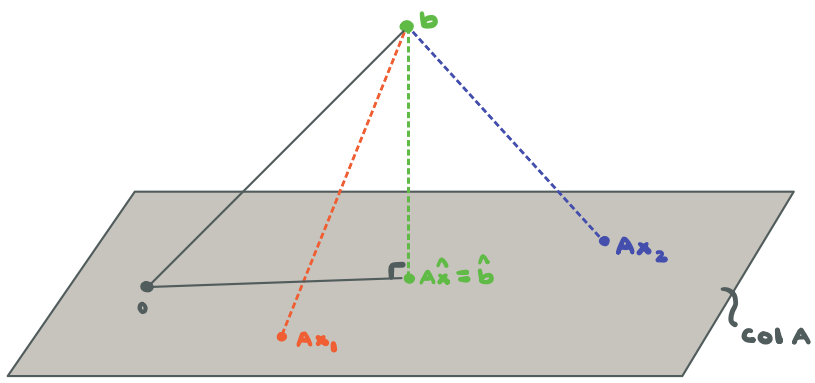
$$||\mathbf{b} - A\hat{\mathbf{x}}|| \leq ||\mathbf{b} - A\mathbf{x}||$$

for all $\mathbf{x} \in \mathbb{R}^n$.

Goal: Find $\hat{\mathbf{x}}$.

Notice that for any $\mathbf{x} \in \mathbb{R}^n$, $A\mathbf{x} \in \operatorname{Col} A$. So, we want to find an $\mathbf{x}$ such that $A\mathbf{x}$ is the closest point in $\operatorname{Col} A$ to $\mathbf{b}$. Applying the best approximation theorem, we have that

$$\hat{\mathbf{b}} = \operatorname{proj}_{\operatorname{Col} A}\mathbf{b}.$$



Since $\hat{\mathbf{b}} \in \operatorname{Col} A$, the equation $A\mathbf{x} = \hat{\mathbf{b}}$ has a solution, i.e. there exists $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$. So, a vector $\hat{\mathbf{x}}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$ if and only if $\hat{\mathbf{x}}$ satisfies $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.

Suppose that $\hat{\mathbf{x}}$ satisfies $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$. By the orthogonal decomposition theorem, $\mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to $\operatorname{Col} A$. This implies that $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to each column of A. Let $\mathbf{a}_j$ be a column of A. Then,

$$\mathbf{a}_j \bullet (\mathbf{b} - A\hat{\mathbf{x}}) = 0 \implies \mathbf{a}_j^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0.$$

We know that $\mathbf{a}_j^T$ is a row of A^T, and so it must be that

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \implies A^T\mathbf{b} - A^TA\hat{\mathbf{x}} = \mathbf{0} \implies A^T\mathbf{b} = A^TA\hat{\mathbf{x}}.$$

So, each least-squares solution of $A\mathbf{x} = \mathbf{b}$ satisfies the equation

$$A^TA\mathbf{x} = A^T\mathbf{b}.$$

The system of equations coming from $A^TA\mathbf{x} = A^T\mathbf{b}$ are the normal equations for $A\mathbf{x} = \mathbf{b}$. A solution is denoted by $\hat{\mathbf{x}}$.

Remark: If $\mathbf{b} \in \operatorname{Col} A$, then $\mathbf{b} = A\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n$. We call $\mathbf{x}$ a "least-squares solution."

Example: Find a least-squares solution of the inconsistent system $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}.$$

Solution: We see that

$$A^TA = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix},$$

$$A^T\mathbf{b} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 16 \\ 17 \end{bmatrix}.$$

So, the equation $A^TA\mathbf{x} = A^T\mathbf{b}$ can be written as

$$\begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 16 \\ 17 \end{bmatrix}.$$

The inverse of A^TA is

$$\frac{1}{36-25} \begin{bmatrix} 6 & -5 \\ -5 & 6 \end{bmatrix} = \begin{bmatrix} 6/11 & -5/11 \\ -5/11 & 6/11 \end{bmatrix}.$$

Therefore,

$$\hat{\mathbf{x}} = \begin{bmatrix} 6/11 & -5/11 \\ -5/11 & 6/11 \end{bmatrix} \begin{bmatrix} 16 \\ 17 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Notice that I could have used an augmented matrix to solve this system (instead of finding the inverse of A^TA).

Note: We can have infinitely many least-squares solutions. However, $\hat{\mathbf{b}}$ is always unique.

Example: Find a least-squares solution of the inconsistent system $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 3 \\ 1 \end{bmatrix}.$$

Solution: We see that

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix},$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ 6 \end{bmatrix}.$$

We want to find $\hat{\mathbf{x}}$ such that $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. So, let's use an augmented matrix.

$$\begin{aligned}
\begin{bmatrix} 4 & 2 & 2 & 10 \\ 2 & 2 & 0 & 4 \\ 2 & 0 & 2 & 6 \end{bmatrix} &\xrightarrow{r_1 \to \frac{r_1}{2}} \begin{bmatrix} 2 & 1 & 1 & 5 \\ 2 & 2 & 0 & 4 \\ 2 & 0 & 2 & 6 \end{bmatrix} \\
&\xrightarrow{r_2 \to r_2 - r_3} \begin{bmatrix} 2 & 1 & 1 & 5 \\ 0 & 2 & -2 & -2 \\ 2 & 0 & 2 & 6 \end{bmatrix} \\
&\xrightarrow{r_2 \to \frac{r_2}{2}} \begin{bmatrix} 2 & 1 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 2 & 0 & 2 & 6 \end{bmatrix} \\
&\xrightarrow{r_3 \to r_3 - r_1} \begin{bmatrix} 2 & 1 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 1 & 1 \end{bmatrix} \\
&\xrightarrow{r_3 \to r_3 + r_2} \begin{bmatrix} 2 & 1 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&\xrightarrow{r_1 \to r_1 - r_2} \begin{bmatrix} 2 & 0 & 2 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&\xrightarrow{r_1 \to \frac{r_1}{2}} \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.
\end{aligned}$$

So, we have that $x_1 + x_3 = 3$ and $x_2 - x_3 = -1$, where x_3 is a free variable. Therefore,

$$\hat{\mathbf{x}} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 - x_3 \\ x_3 - 1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

for any constant x_3.

Theorem: Let A be an $m \times n$ matrix. The following are all equivalent.

- The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b} \in \mathbb{R}^m$.
- The columns of A are linearly independent.
- The matrix $A^T A$ is invertible.

If these statements are true, then the least-squares solution is

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}.$$

Theorem: Let A be a $m \times n$ matrix with linearly independent columns. Suppose we have a QR factorization of A, namely $A = QR$. Then, for each $\mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution, given by $\hat{\mathbf{x}} = R^{-1} Q^T \mathbf{b}$.